

Supercomputing Time Allocation Request

Apex Petascale Computing Facility | Nexa National HPC Allocation Program, Large-Scale Track

Direct Numerical Simulation of Extinction and Reignition Dynamics in Hypersonic Scramjet Combustors

Request ID: NEXA-2026-LS-0847 **Cycle:** 14 **Track:** Large-Scale Submitted 2026-07-23
(>1M SU)

Principal Investigator: Dr. Elena Castillo
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Requested period: 2026-10-01 – 2027-09-30
(12 months)

Co-Investigator: Dr. Rajiv Patel — Computational Combustion Group, Synapse National Laboratory

Target resource: Apex Phase-II GPU Partition, Nimbus Computing Center



4.20M
GPU-hours (H100)
requested



1.80M
CPU core-hours
(pre/post)



725 TB
Peak scratch +
archive storage



1024
Max. GPU concurrency demonstrated

Executive Summary

We request **4,200,000 GPU-hours** on the Apex Phase-II partition (Lumina H100, Nimbus Computing Center) to conduct the first suite of wall-resolved direct numerical simulations (DNS) of local extinction and reignition in a Mach 6.5 hydrogen-fueled scramjet combustor. These flows govern the practical operability limits of hypersonic air-breathing propulsion, yet no existing simulation campaign has resolved the coupled turbulence–chemistry interaction at engine-relevant Reynolds and Damköhler numbers. Our solver, **VertexCFD**, has been benchmarked to 1,024 H100 GPUs at 74% parallel efficiency (Section 2.2) and has consumed a prior Apex Phase-I allocation productively (Section 5). The requested allocation funds six production DNS configurations spanning the extinction/reignition boundary, together with the grid-convergence and reduced-mechanism validation studies required to certify each result.

1 Project Summary

Scramjet combustors operate in a narrow, poorly characterized envelope between blow-out and thermal choking. Small perturbations in fuel–air mixing — a bow-shock/boundary-layer interaction, an injector wake, a transient equivalence-ratio excursion — can locally extinguish the flame; whether it reignites downstream determines whether the vehicle sustains thrust or loses the engine entirely. Reduced-order models (flamelet, PDF-transport) are calibrated against subsonic or low-Damköhler data and are known to mispredict extinction onset by a factor of two or more at hypersonic conditions. The only way to build a trustworthy predictive model is to resolve the full range of turbulent and chemical scales directly, without closure assumptions, in geometry-accurate combustor sections. That computation is only tractable at the scale of a national allocation.